Connecting Networks

AWS Academy Cloud Architecting

Introduction

Connecting Networks

Module objectives

This module prepares you to do the following:

- Describe how to connect an on-premises network to the Amazon Web Services (AWS) Cloud.
- Describe how to connect multiple virtual private clouds (VPCs) in the AWS Cloud.
- Connect VPCs in the AWS Cloud by using VPC peering.
- Describe how to scale VPCs in the AWS Cloud.
- Use the AWS Well-Architected Framework principles when connecting networks.

Module overview

Presentation sections

- Scaling your VPC network with AWS Transit Gateway
- Connecting VPCs in AWS with VPC peering
- Connecting to your remote network with AWS Site-to-Site VPN
- Connecting to your remote network with AWS Direct Connect
- Applying AWS Well-Architected Framework to network connectivity

Activity

- Configure AWS Transit Gateway Routes

Knowledge checks

- 10-question knowledge check
- Sample exam question

Hands-on labs in this module

Guided lab

- Creating a VPC Peering Connection

As a cloud architect designing connected networks:

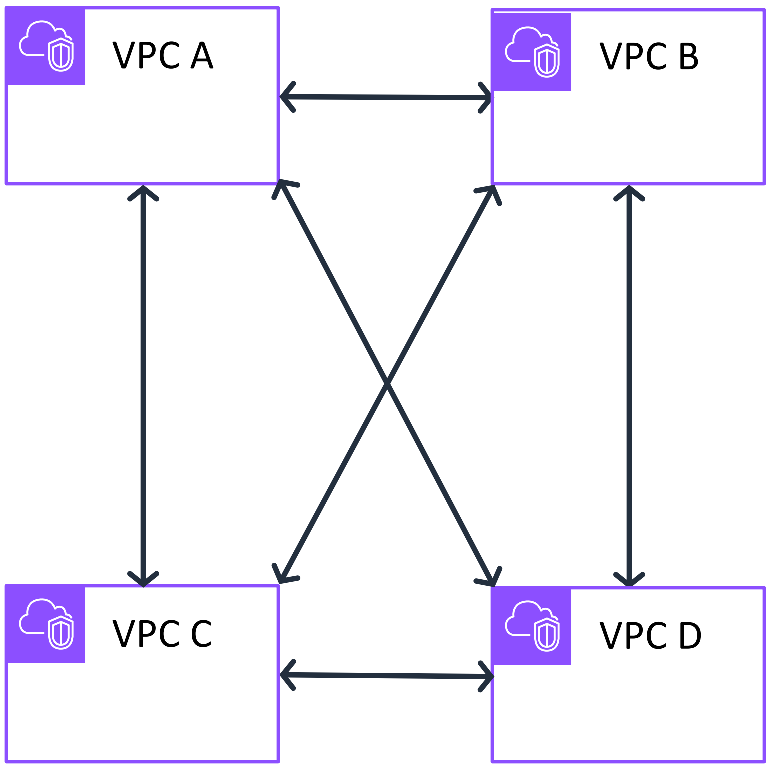
- I need to design for failover capability and adequate bandwidth to accommodate applications deployed on premises, in the cloud, or as a hybrid solution, so that my networks perform as needed.
- I need to choose network components that optimize performance and reduce data transfer costs between networks, so that I can maximize the business value.
- I need to protect data in transit between networks, so that I can meet data security compliance requirements.

Scaling your VPC network with AWS Transit Gateway

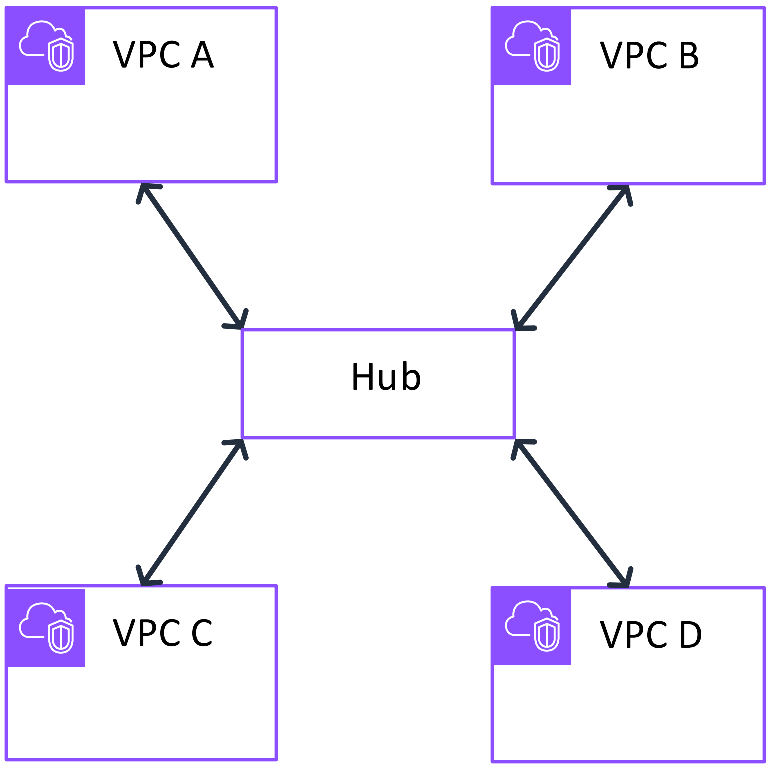
Connecting Networks

Network design with multiple VPCs

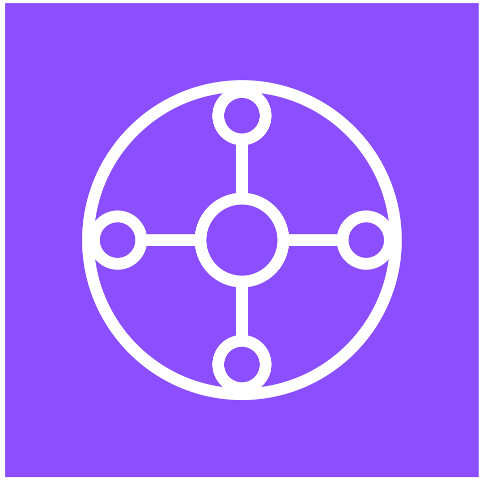
Full mesh architecture



Hub-and-spoke architecture



AWS Transit Gateway



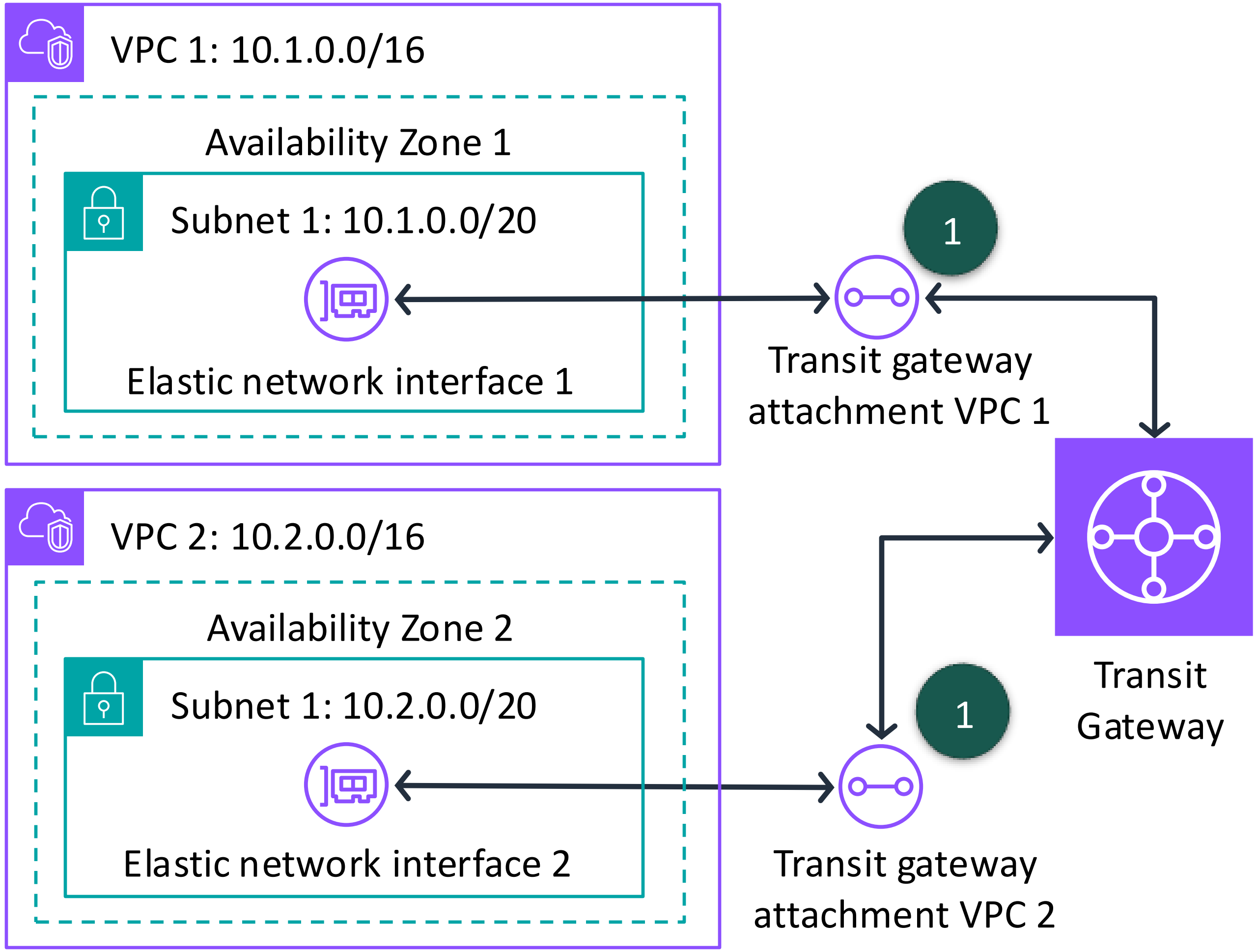
Transit Gateway

- Is a centralized, Regional router to connect VPCs and on-premises networks based on hub-and-spoke architecture
- Is a managed AWS service that automatically scales based on the volume of network traffic
- Can be peered with other transit gateways in other AWS Regions and AWS accounts
- Incurs cost charges based on the number of connections and amount of traffic throughput
- Has a Transit Gateway Flow Logs feature to publish transit gateway traffic logs

Centralized VPC routing pattern

VPC 1 route table	
Destination	**Target**
10.1.0.0/16	local
10.0.0.0/8	Transit gateway ID

VPC 2 route table	
Destination	**Target**
10.2.0.0/16	local
10.0.0.0/8	Transit gateway ID



Transit gateway route table	
Destination	**Target**
10.1.0.0/16	Transit gateway attachment VPC 1 ID
10.2.0.0/16	Transit gateway attachment VPC 2 ID

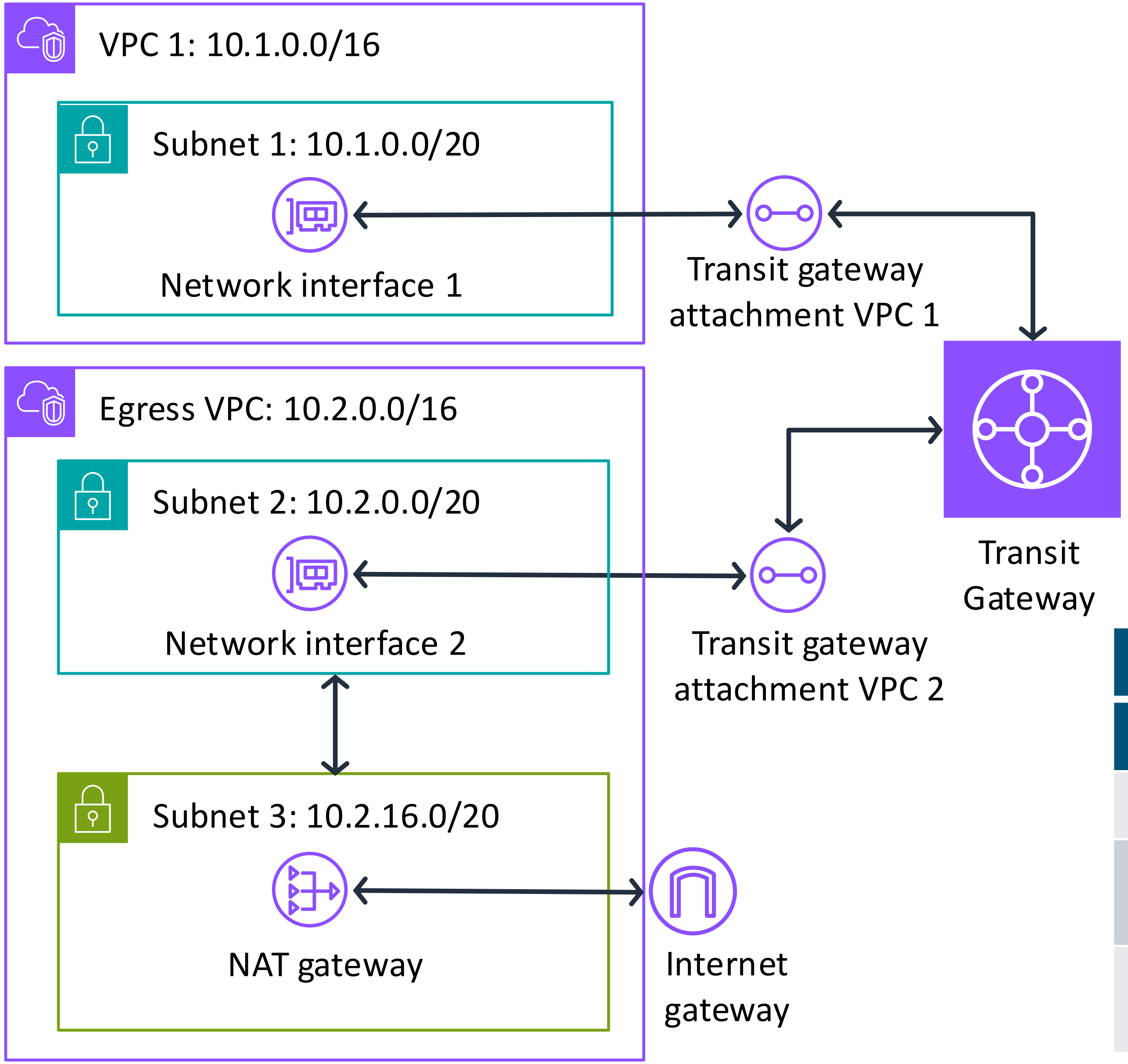
Centralized routing pattern for outbound traffic

VPC 1 route table

Destination	Target
10.1.0.0/16	local
0.0.0.0/0	Transit gateway ID

Subnet 2 route table

Destination	Target
10.1.0.0/16	local
0.0.0.0/0	NAT gateway ID



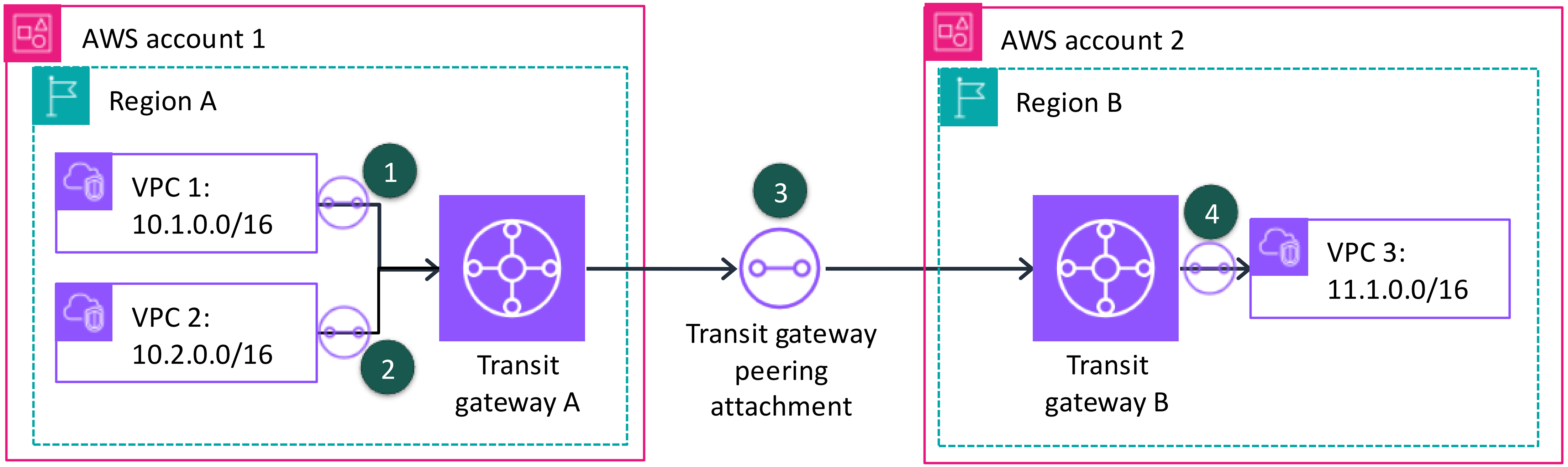
Transit gateway route table

Destination	Target
10.1.0.0/16	Transit gateway attachment VPC 1 ID
10.2.0.0/16	Transit gateway attachment VPC 2 ID

Egress VPC route table

Destination	Target
10.2.0.0/16	local
10.1.0.0/16	Transit gateway ID
0.0.0.0/0	Internet gateway ID

Transit gateway peering



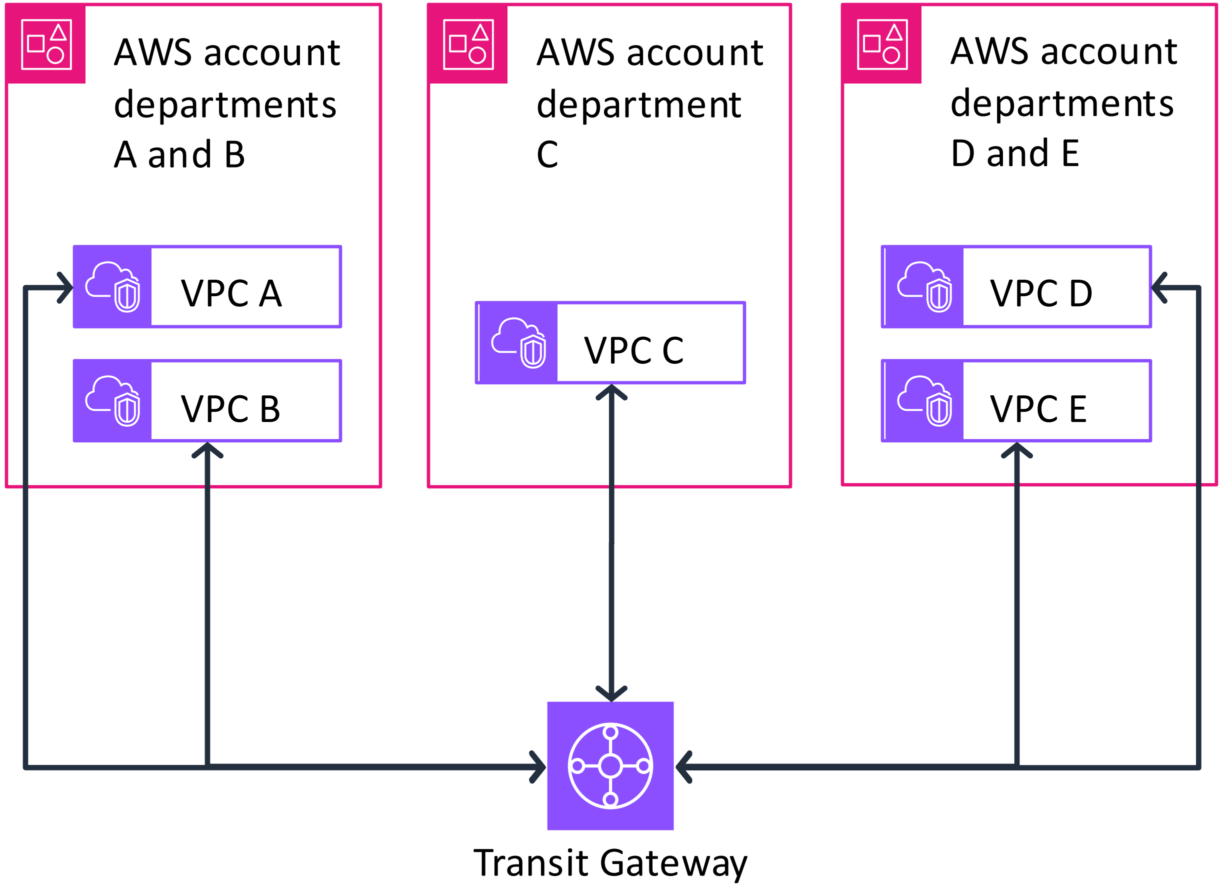
Transit gateway A route table		
Destination	Target	
10.1.0.0/16	Transit gateway attachment VPC 1 ID	1
10.2.0.0/16	Transit gateway attachment VPC 2 ID	2
11.1.0.0/16	Transit gateway peering attachment ID	3

Transit gateway B route table		
Destination	Target	
11.1.0.0/16	Transit gateway attachment VPC 3 ID	4
10.1.0.0/16	Transit gateway peering attachment ID	3
10.2.0.0/16	Transit gateway peering attachment ID	3

Company group of departments use case

Scenario:

A company has multiple IT departments, each with their own VPC. Some VPCs are located within the same AWS account, and others in a different AWS account. All the VPCs should be peered together to let the IT departments have full access to each others' resources. The company is considering adding accounts of other business groups in the future.



Activity: Configure AWS Transit Gateway Routes

- Configure routes for five VPCs that you want to connect to each other through Transit Gateway.
- Configure routes in the VPC route tables.
- Configure routes in the transit gateway route table.

Configure the VPC route tables

Which routes are necessary to add to each of the VPC route tables to provide full connectivity between all five VPCs?

VPC ID	VPC CIDR	Transit gateway VPC attachment ID	VPC route table ID	VPC route table destination	VPC route table target
vpc-a	10.1.0.0/16	tgw-attach-vpc-a	rtb-vpc-a	?	?
vpc-b	10.2.0.0/16	tgw-attach-vpc-b	rtb-vpc-a	?	?
vpc-c	10.3.0.0/16	tgw-attach-vpc-c	rtb-vpc-a	?	?
vpc-d	10.4.0.0/16	tgw-attach-vpc-d	rtb-vpc-a	?	?
vpc-e	10.5.0.0/16	tgw-attach-vpc-e	rtb-vpc-a	?	?

Configure the transit gateway tgw-1 route table

Which routes are necessary to add to the transit gateway tgw-1 route table to provide full connectivity between all five VPCs?

Transit gateway route table destination	Transit gateway route table target
?	?
?	?
?	?
?	?
?	?

Solution: VPC route tables

VPC ID	VPC CIDR	Transit gateway VPC attachment ID	VPC route table ID	VPC route table destination	VPC route table target
vpc-a	10.1.0.0/16	tgw-attach-vpc-a	rtb-vpc-a	10.0.0.0/8	tgw-1
vpc-b	10.2.0.0/16	tgw-attach-vpc-b	rtb-vpc-a	10.0.0.0/8	tgw-1
vpc-c	10.3.0.0/16	tgw-attach-vpc-c	rtb-vpc-a	10.0.0.0/8	tgw-1
vpc-d	10.4.0.0/16	tgw-attach-vpc-d	rtb-vpc-a	10.0.0.0/8	tgw-1
vpc-e	10.5.0.0/16	tgw-attach-vpc-e	rtb-vpc-a	10.0.0.0/8	tgw-1

Solution: Transit gateway tgw-1 route table

Transit gateway route table destination	Transit gateway route table target
10.1.0.0/16	tgw-attach-vpc-a
10.2.0.0/16	tgw-attach-vpc-b
10.3.0.0/16	tgw-attach-vpc-c
10.4.0.0/16	tgw-attach-vpc-d
10.5.0.0/16	tgw-attach-vpc-e

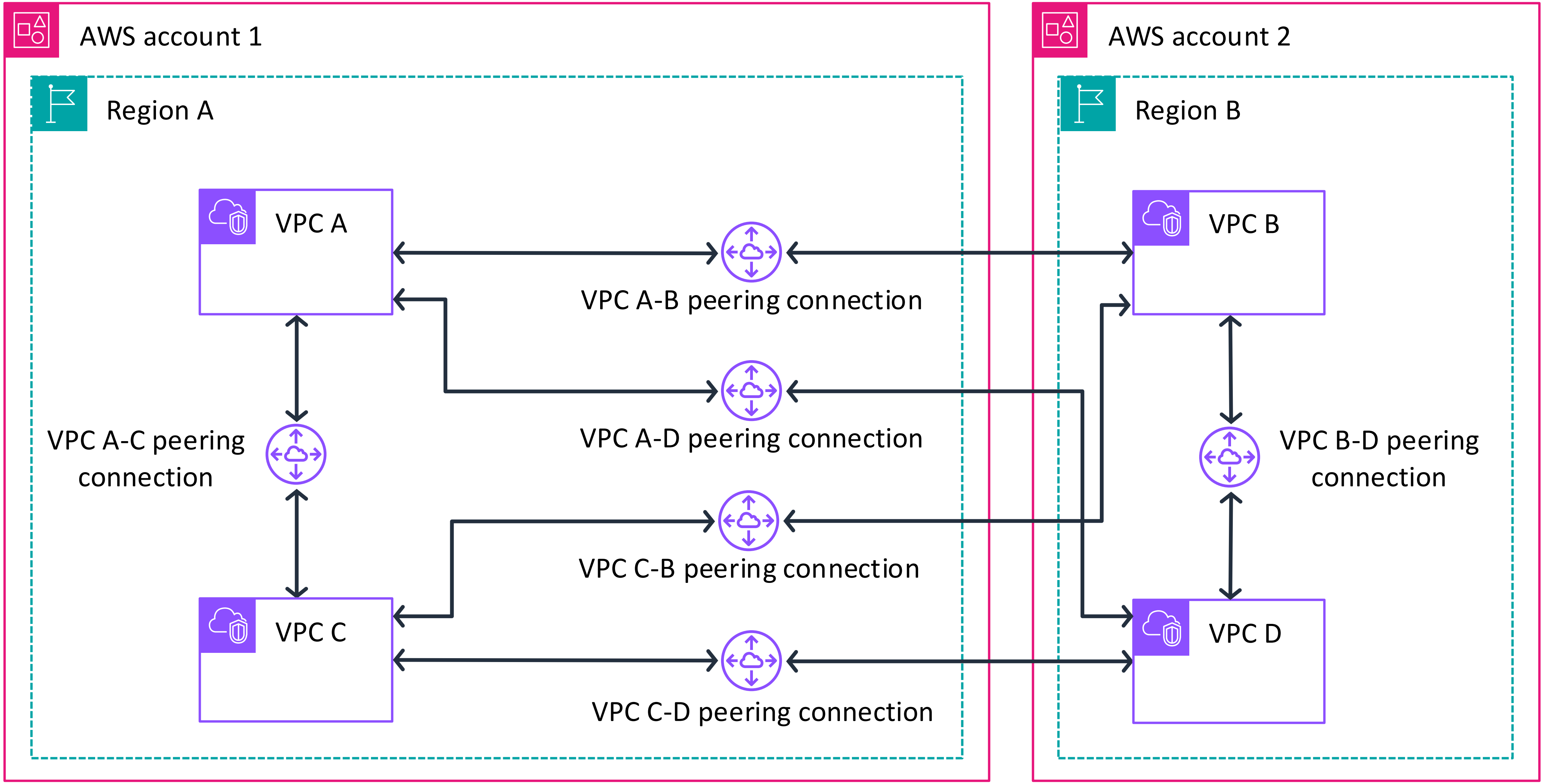
Key takeaways: Scaling your VPC network with Transit Gateway

- Transit Gateway is a centralized Regional router to connect VPCs.
- Transit Gateway can be peered with other transit gateways in other AWS Regions and AWS accounts.
- Transit Gateway supports thousands of attachments.
- Transit Gateway charges per hour for the number of connections that you make to the transit gateway and the amount of traffic that flows through the transit gateway.

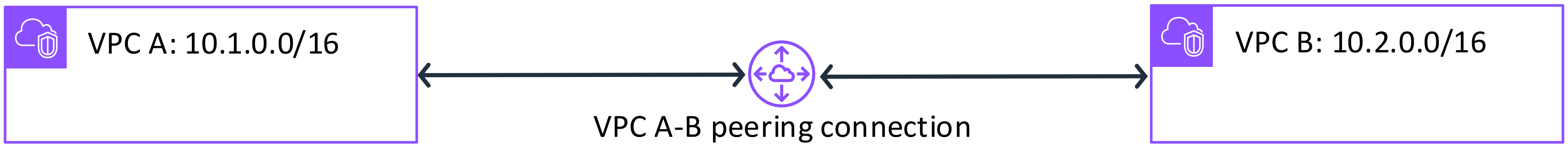
Connecting VPCs in AWS with VPC peering

Connecting Networks

Mesh architecture using VPC peering



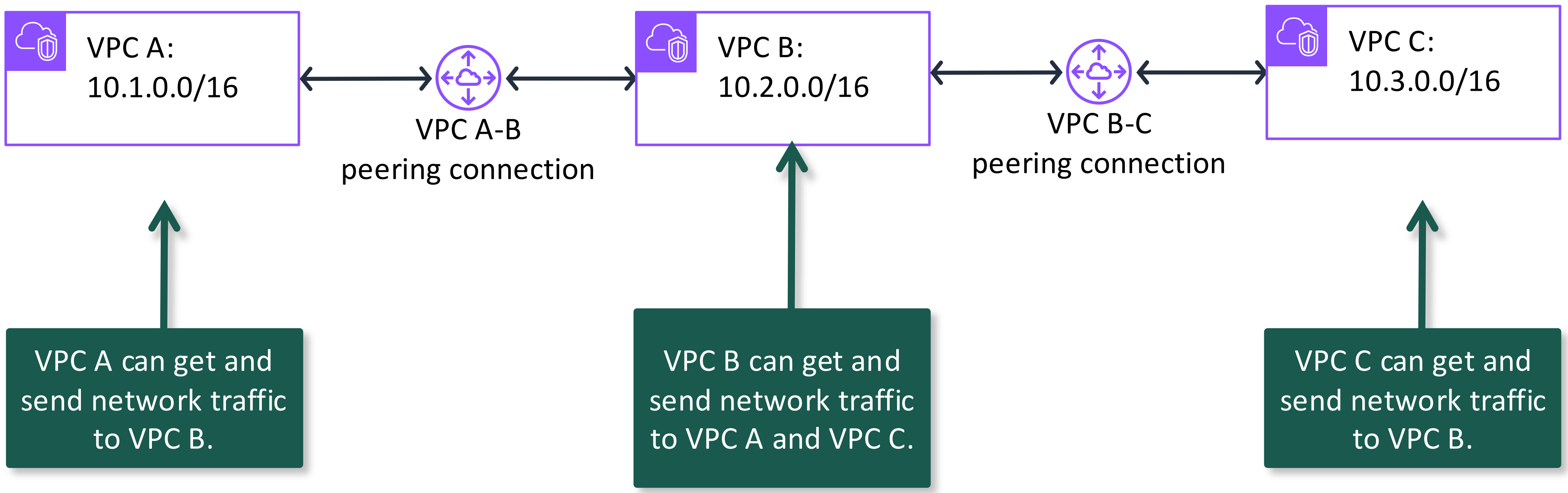
Establishing **VPC peering**



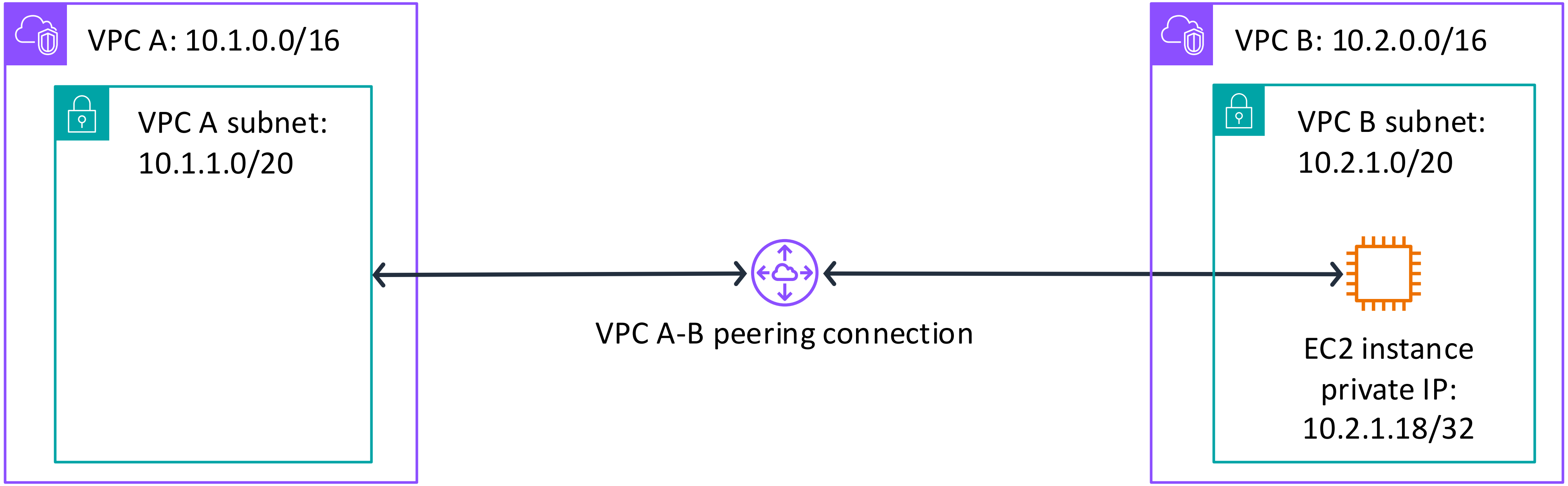
VPC A route table	
Destination	**Target**
10.1.0.0/16	local
10.2.0.0/16	VPC A-B peering connection ID

VPC B route table	
Destination	**Target**
10.2.0.0/16	local
10.1.0.0/16	VPC A-B peering connection ID

VPC peering does not support transitive peering



VPC peering configurations with specific routes

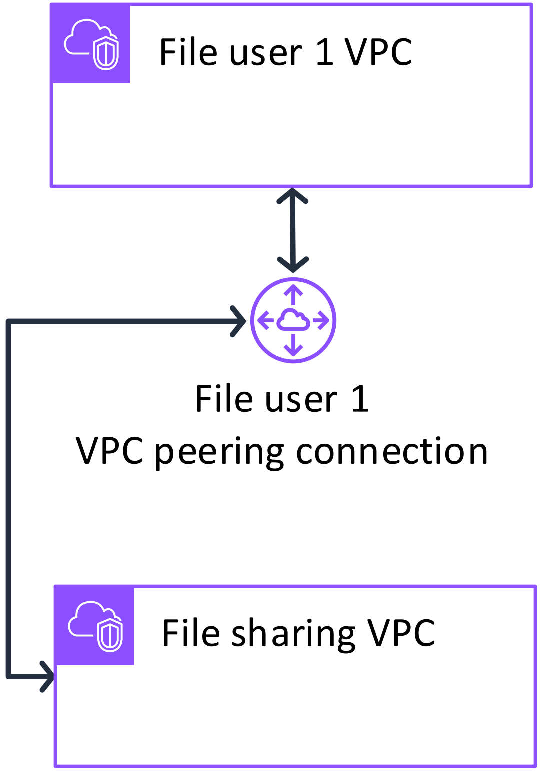


VPC A subnet route table	
Destination	**Target**
10.1.0.0/16	local
10.2.1.18/32	VPC A-B peering connection ID

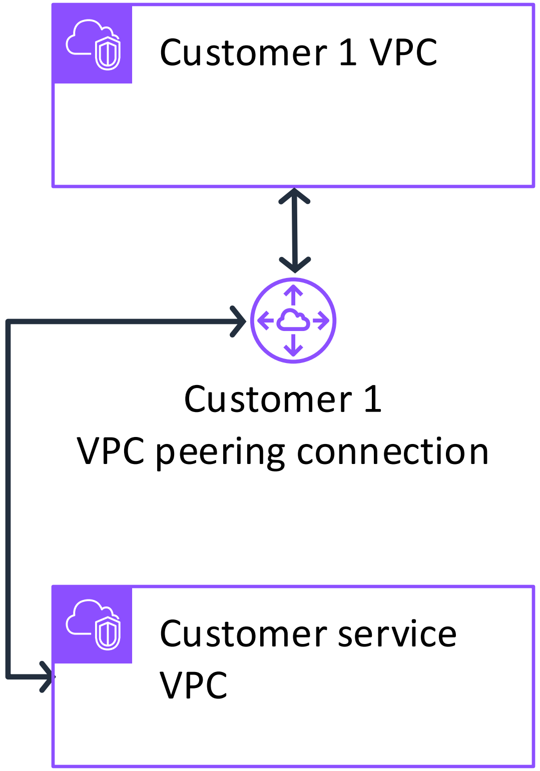
VPC B subnet route table	
Destination	**Target**
10.2.0.0/16	local
10.1.1.0/20	VPC A-B peering connection ID

Use cases: Peering to one VPC to access centralized resources

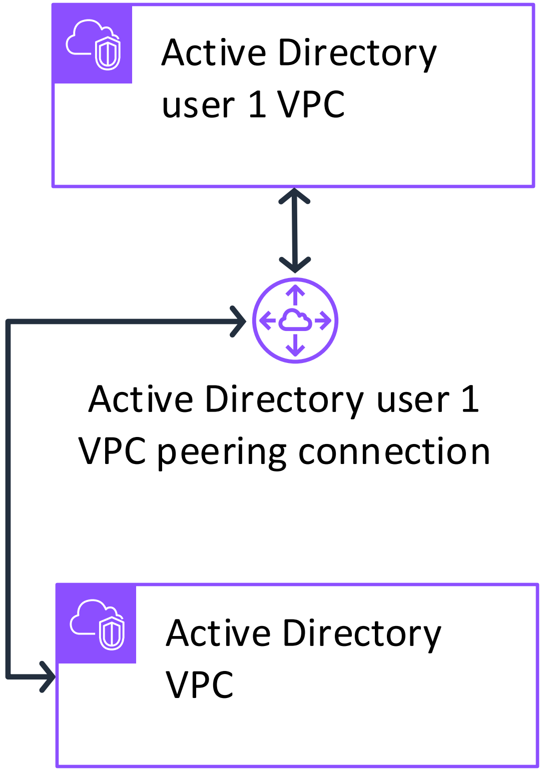
1. File sharing VPC



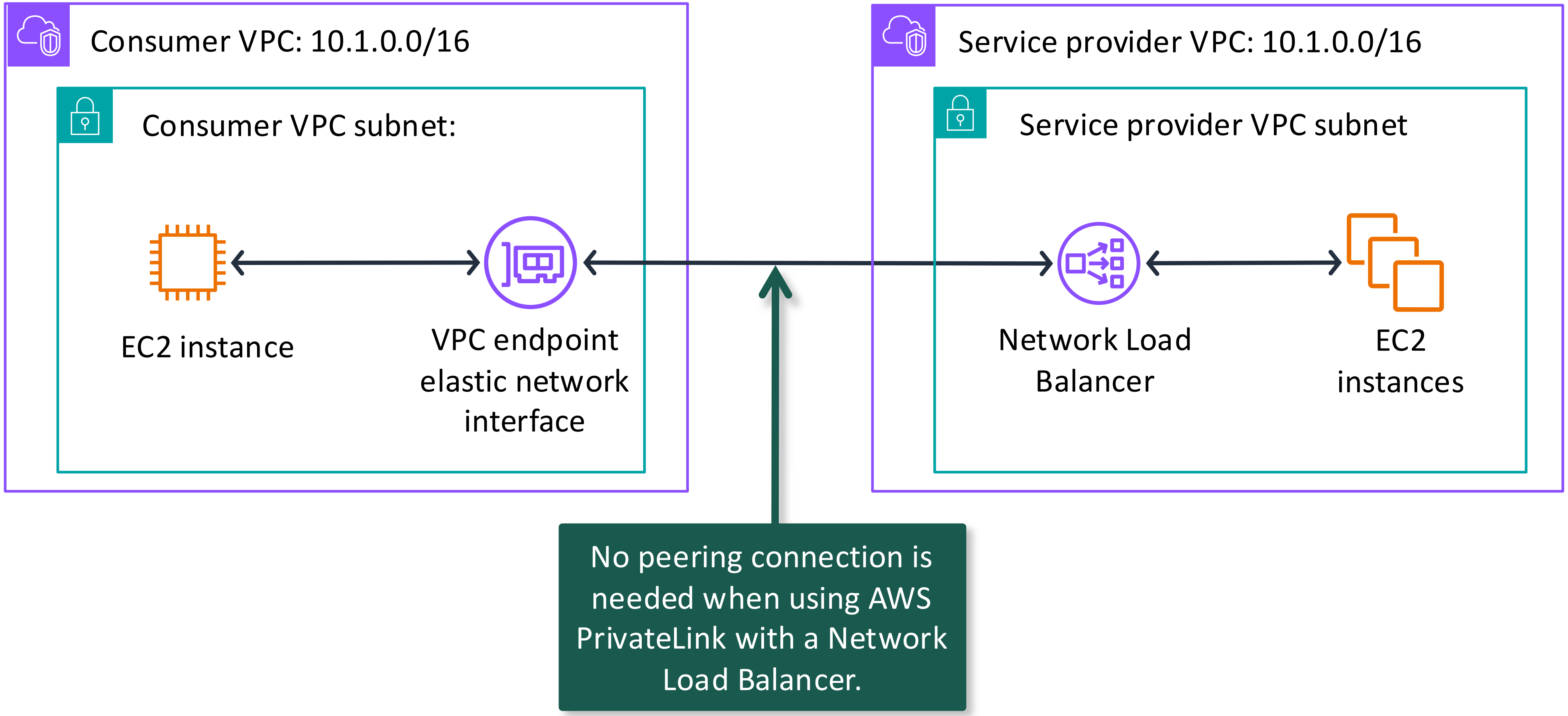
2. Customer shared VPC



3. Active Directory VPC



VPC connectivity with AWS PrivateLink architecture



Key takeaways: Connecting VPCs in AWS with VPC peering

- VPC peering establishes a one-to-one peering networking connection between two VPCs to provide private network traffic routes.
- VPC peering does not incur costs, but transferring data across Availability Zones and Regions does.
- VPC peering can provide network traffic flow between different AWS accounts and AWS Regions.
- VPC peering does not support transitive VPC peering relationships.
- If VPC Classless Inter-Domain Routing (CIDR) blocks overlap, use PrivateLink with a Network Load Balancer to establish VPC connectivity.

Guided lab: Creating a VPC Peering Connection (VPC Peering Lab)

VPC Peering Lab tasks

- In this lab, you will perform the following main tasks:
  - Create a peering connection between two VPCs.
  - Configure route tables to send traffic to the peering connection.
  - Test the peering connection.
- Open your lab environment to start the lab and find additional details about the tasks that you will perform.

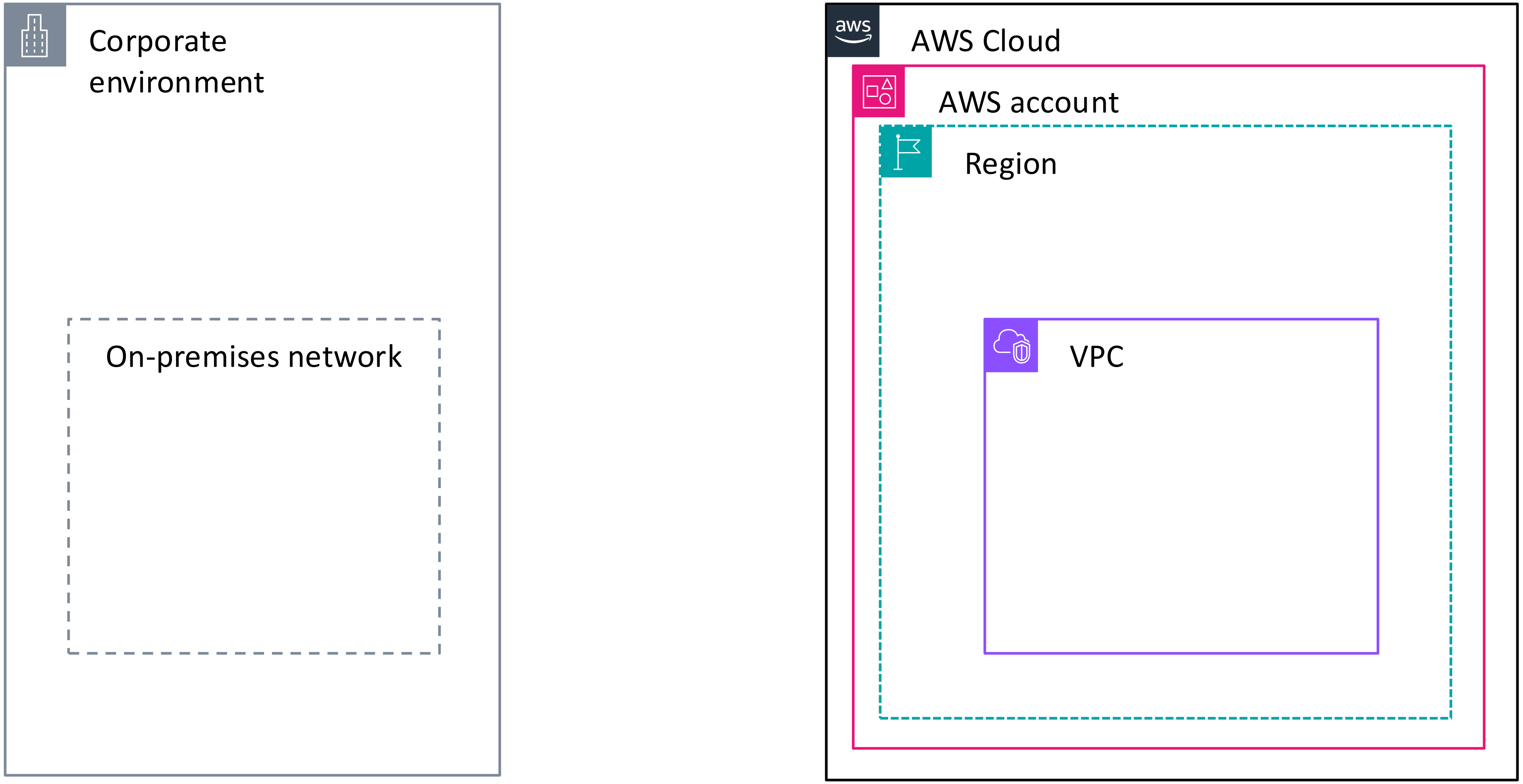
Debrief: VPC peering lab

- What did you learn about the VPC peering connection process?
- How were you able to confirm that the VPC peering worked?

Connecting to your remote network with AWS Site-to-Site VPN

Connecting Networks

Connecting on-premises environments to a VPC

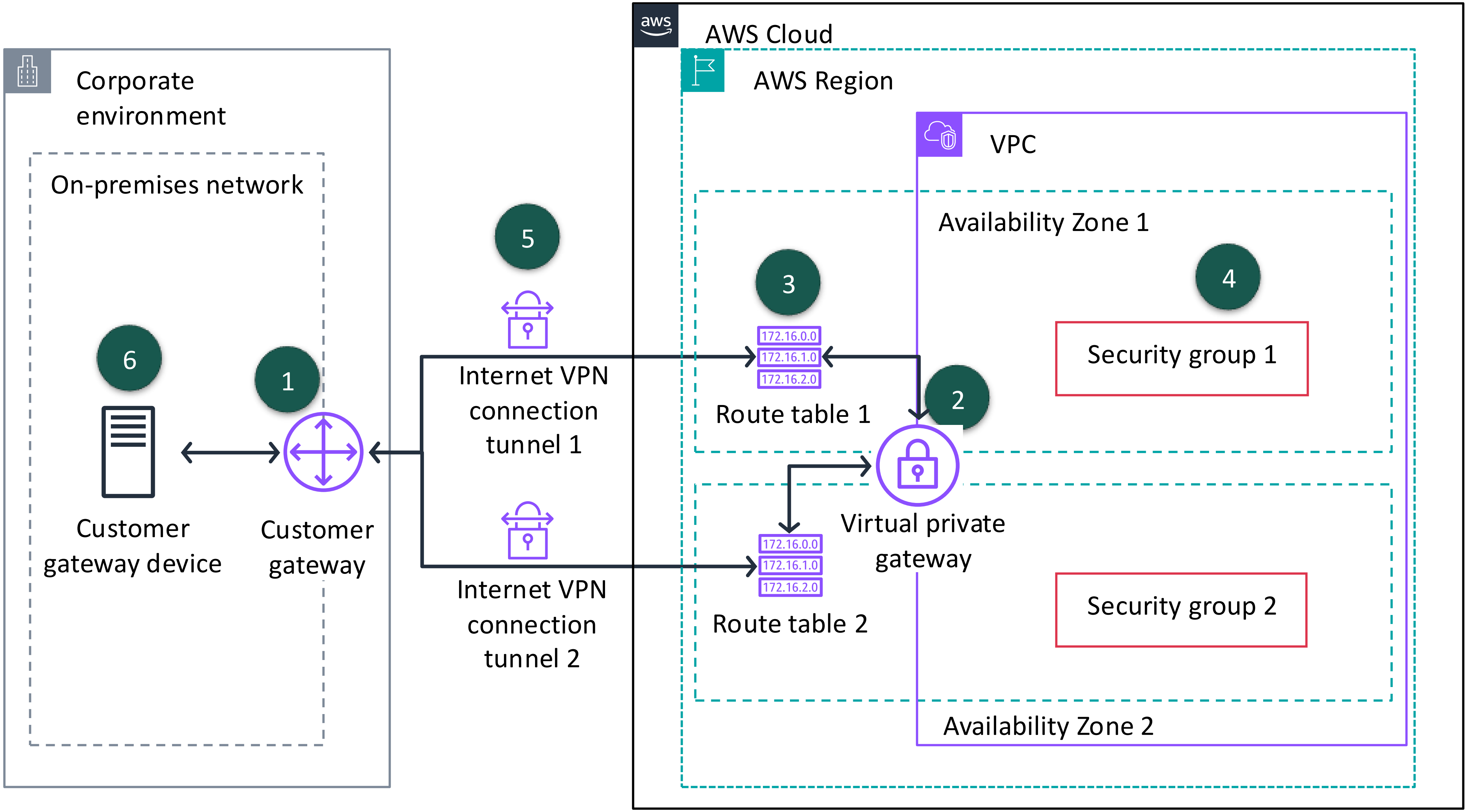


AWS Site-to-Site VPN

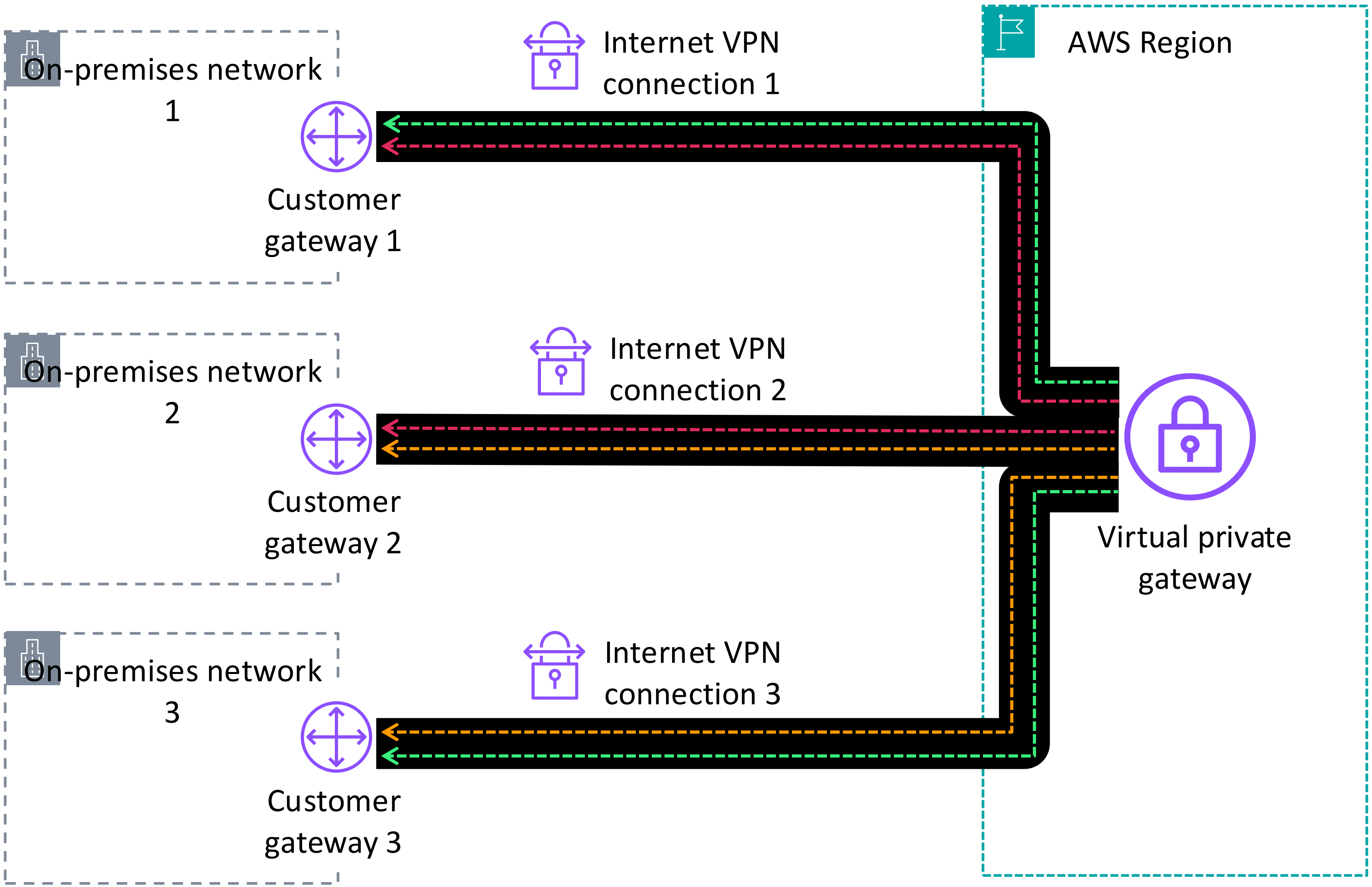
Site-to-Site VPN

- Creates a secure connection between an on-premises customer gateway and AWS virtual private gateway (or transit gateway) for VPC access
- Creates two encrypted IPsec VPN tunnels for each connection across multiple Availability Zones
- Charges for each VPN connection-hour

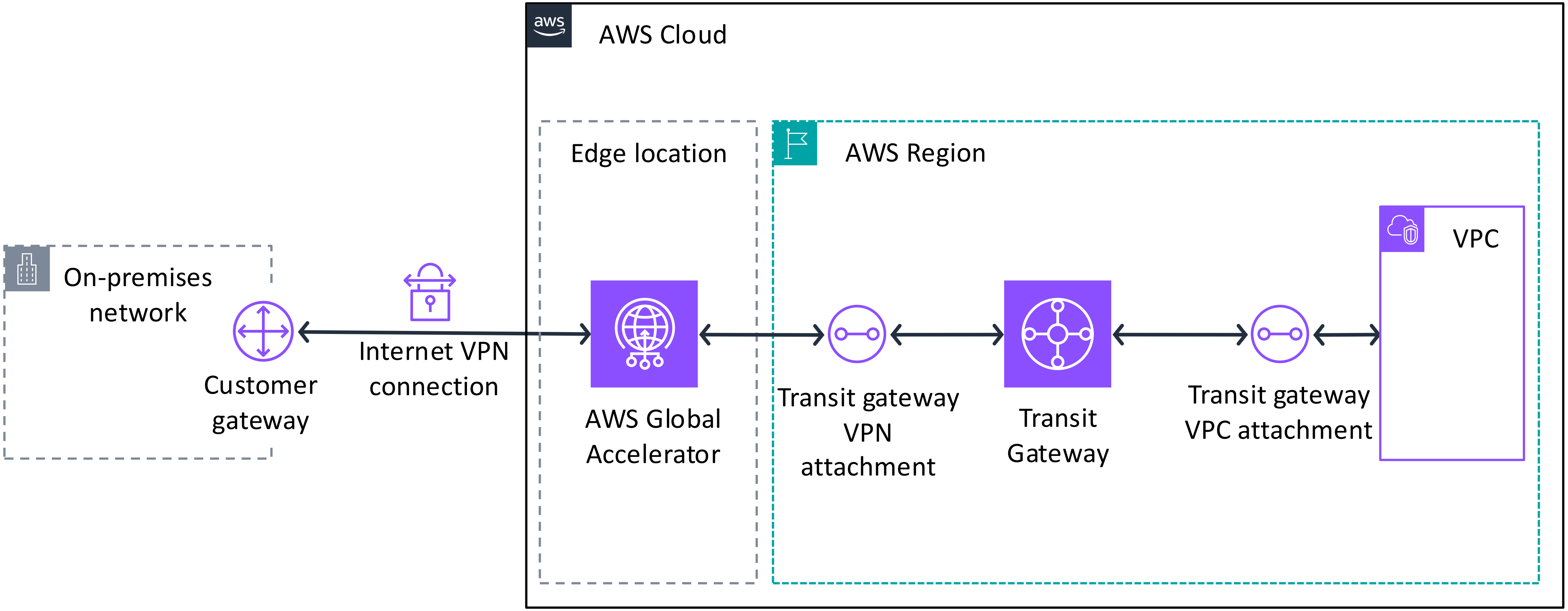
Creating a Site-to-Site VPN connection



Connecting multiple on-premises networks by using AWS VPN CloudHub

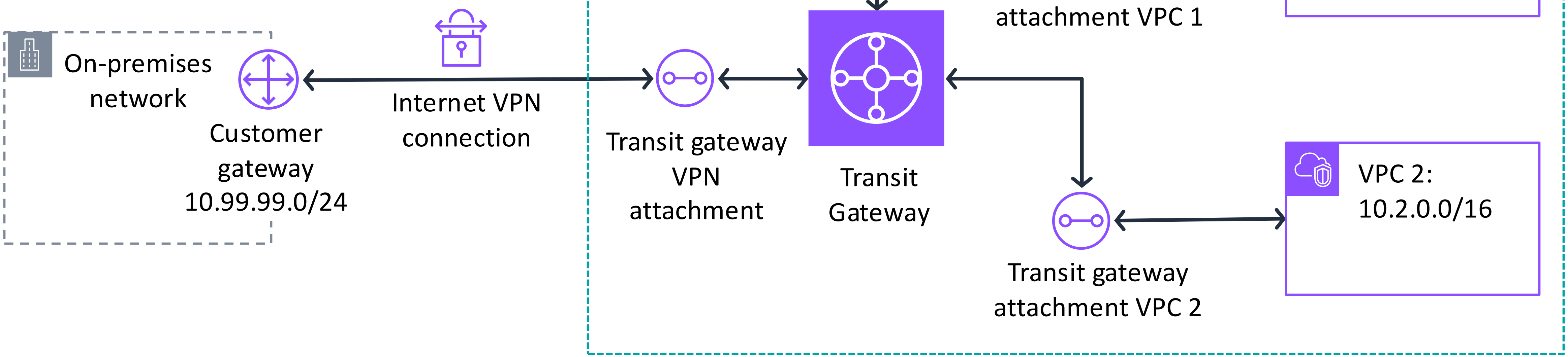


Accelerating Site-to-site VPN connections



Isolating VPCs with full VPN access by using Transit Gateway

On-premises route table	
Destination	**Target**
10.X.0.0/16	local
0.0.0.0/0	Transit gateway ID



Transit gateway VPN route table	
Destination	**Target**
10.1.0.0/16	Transit gateway attachment VPC 1 ID
10.2.0.0/16	Transit gateway attachment VPC 2 ID

Transit gateway VPC route table	
Destination	**Target**
10.99.99.0/24	Transit gateway VPN attachment ID

Key takeaways: Connecting to your remote network with Site-to-Site VPN

- Site-to-Site VPN creates a secure connection between an on-premises customer gateway and an AWS virtual private gateway (or transit gateway) for VPC access.
- You can connect multiple on-premises networks to a single virtual private gateway.
- Use Global Accelerator to accelerate Site-to-Site VPN connections.
- Configure multiple Transit Gateway routing tables to isolate VPCs to provide full VPN access.

Connecting to your remote network with AWS Direct Connect

Connecting Networks

AWS Direct Connect

Direct Connect

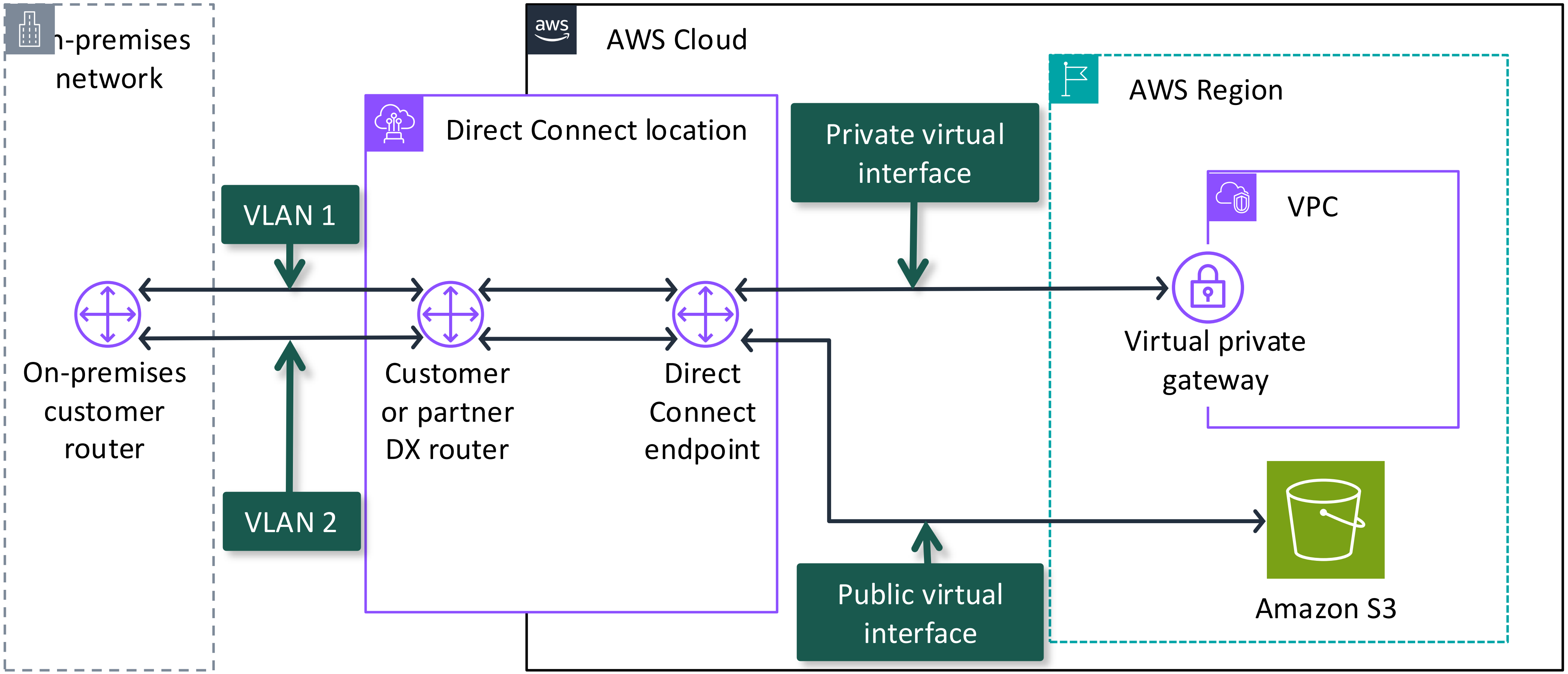
- Is a dedicated, private, virtual local area network (VLAN) connection that extends on-premises network to include AWS resources
- Provides a consistent network experience with predictable performance and increased bandwidth and throughput

Direct Connect use cases

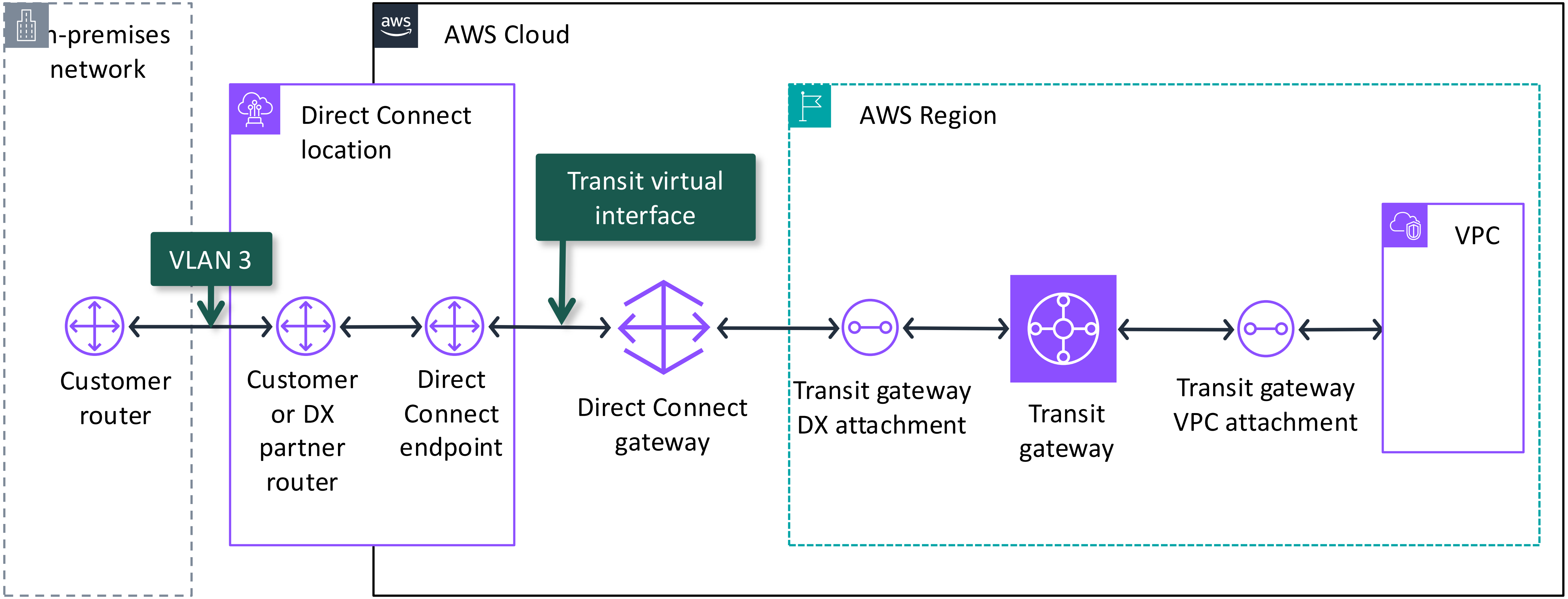



Hybrid environments	Large datasets	Predictable network performance	Security and compliance requirements

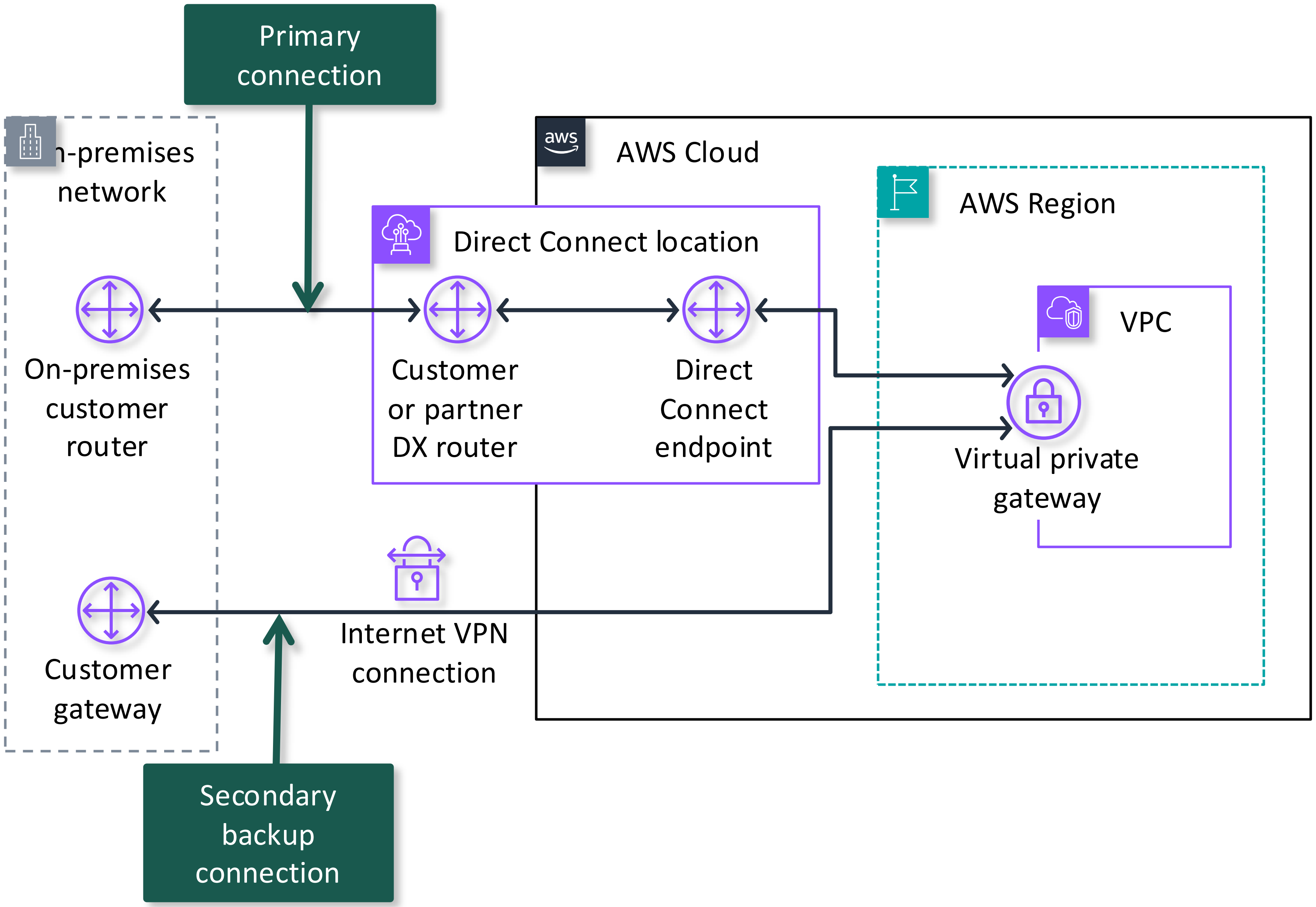
Extend an on-premises network to AWS by using Direct Connect



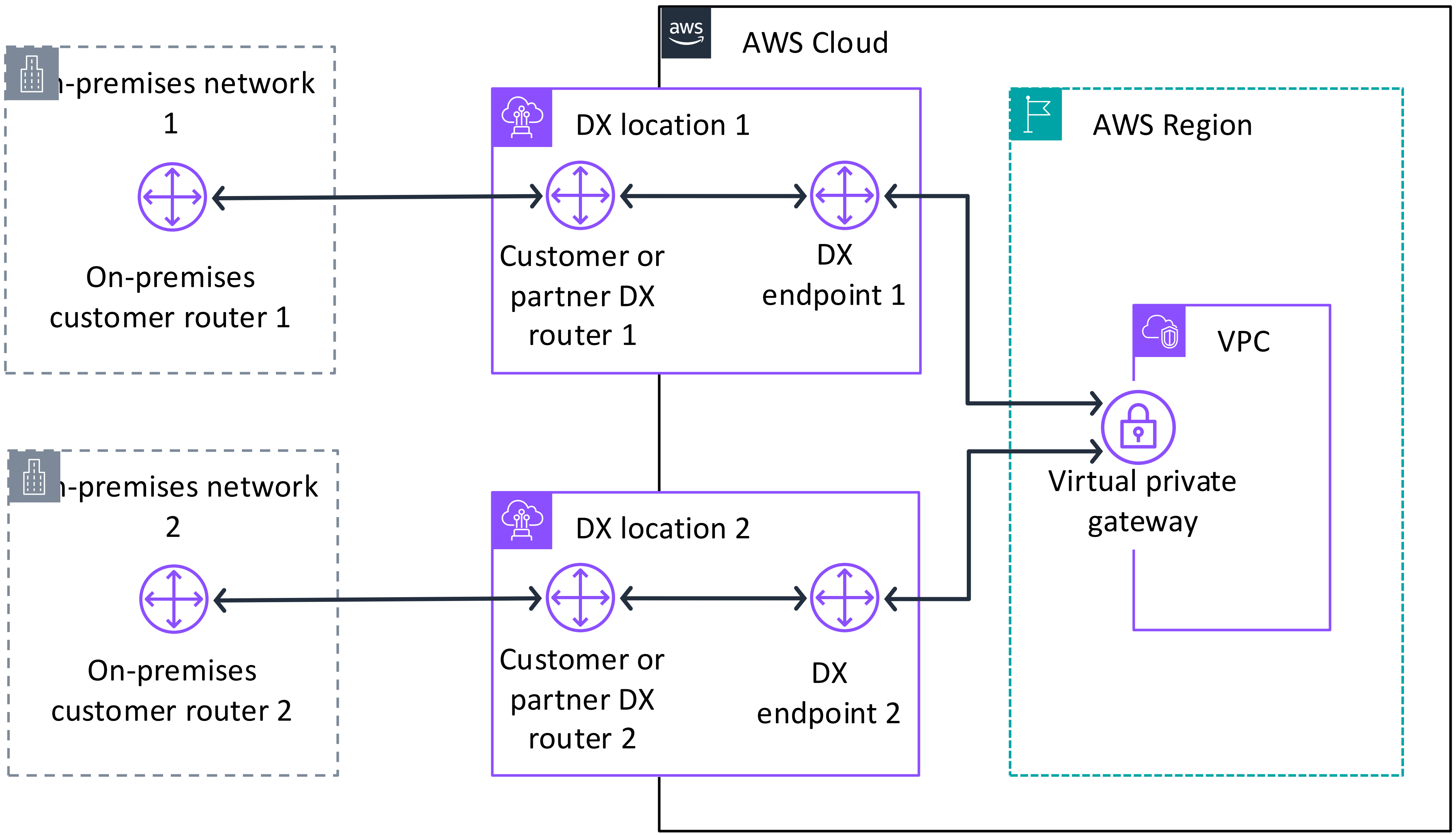
Direct Connect using Transit Gateway



High availability Direct Connect and backup VPN use case



High resiliency with multiple Direct Connect locations



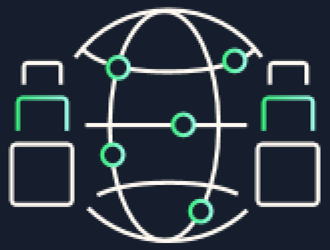
Key takeaways: Connecting to your remote network with Direct Connect

- Direct Connect is a dedicated, private, VLAN connection that extends an on-premises network to include AWS resources.
  - Use a private virtual interface to connect a Direct Connect location to a virtual private gateway.
  - Use a public virtual interface to connect a Direct Connect location to supported AWS services.
  - Use a transit virtual interface to connect a Direct Connect location to a transit gateway through a Direct Connect gateway.
- Make your network highly available by using Direct Connect as a primary connection and a VPN as a backup connection.
- Make your network highly resilient by connecting from multiple on-premises networks to AWS by using multiple Direct Connect locations.

Applying AWS Well-Architected Framework principles to network connectivity

Connecting Networks

AWS Well-Architected Framework network pillars



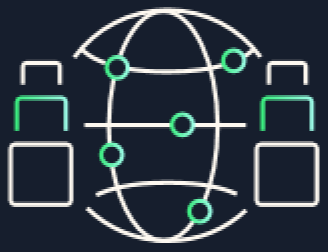
Reliability

Security

Performance Efficiency

Cost Optimization

Best practice approach: Foundations - plan your network topology



Reliability

Best practices

Provision redundant connectivity between private networks in the cloud and in on-premises environments.

Prefer hub-and-spoke topologies over many-to-many mesh.

Best practice approach: Infrastructure protection – protecting networks

Best practice

Control traffic at all layers.

Best practice approach: Data protection – protecting data in transit

Best practice

Authenticate network communications.

Enforce encryption in transit.

Best practice approach: Selection – network architecture selection

Performance Efficiency

Best practices

Choose appropriately sized dedicated connectivity or VPN for hybrid workloads.

Choose your workload's location based on network requirements.

Best practice approach: Cost effective resources – plan for data transfer

Cost Optimization

Best practices

Select components to optimize data transfer cost.

Implement services to reduce data transfer costs.

Key takeaways: Applying Well-Architected Framework principles to your network

- Provision redundant connectivity between private networks in the cloud and in on-premises environments.
- Prefer hub-and-spoke topologies over many-to-many mesh.
- Control traffic at all layers.
- Authenticate network communications.
- Enforce data encryption in transit.
- Select components to optimize and reduce data transfer cost.
- Choose appropriately sized dedicated connectivity or VPN for hybrid workloads.
- Choose your workload’s location based on network requirements.

Module wrap-up

Connecting Networks

Module summary

This module prepared you to do the following:

- Describe how to connect an on-premises network to the AWS Cloud.
- Describe how to connect multiple VPCs in the AWS Cloud.
- Connect VPCs in the AWS Cloud by using VPC peering.
- Describe how to scale VPCs in the AWS Cloud.
- Use the AWS Well-Architected Framework principles when connecting networks.

Considerations for the café

- Discuss how you as a cloud architect might advise the café based on the key cloud architect concerns presented at the start of this module.

Module knowledge check

- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

An application running on Amazon EC2 instances in a virtual private cloud (VPC) processes sensitive information stored on Amazon S3. The information is accessed by using an Amazon S3 public Regional endpoint over the internet. The security team is concerned that the internet connectivity to Amazon S3 is a security risk. Which solution will resolve the security concern with the most efficient network route?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- EC2 instances in a VPC
- Sensitive information
- Internet connectivity to Amazon S3 is a security risk
- Most efficient network route

Sample exam question: Response choices

An application running on Amazon EC2 instances in a virtual private cloud (VPC) processes sensitive information stored on Amazon S3. The information is accessed by using an Amazon S3 public regional endpoint over the internet. The security team is concerned that the internet connectivity to Amazon S3 is a security risk. Which solution will resolve the security concern with the most efficient network route?

Choice	Response
A	Access the data through an internet gateway.
B	Access the data through a virtual private network (VPN) connection.
C	Access the data through a NAT gateway.
D	Access the data through a VPC endpoint for Amazon S3.

Sample exam question: Answer

The answer is D.

Choice	Response
A	
B	
C	
D	Access the data through a VPC endpoint for Amazon S3.

Thank you

Corrections, feedback, or other questions?
Contact us at https://support.aws.amazon.com/#/contacts/aws-academy.